VIDEO GAME CREATION

Lesson 9 - Core Mechanics III: On the Scoreboard

Semester 1, Week 9 - Score Tracking + Real On-Screen UI

- **Objectives:** Add score tracking to the existing hit-detection code path; build a persistent, always-visible HUD using each engine's real UI tool; explain the difference between push-based and poll-based display updates; recognize that some engines' Lesson 6-8 text tools were already the 'real' UI all along.
- **IGDA Tie-In:** Programming - UI/UX implementation, state display, event-driven vs. polling architecture (intermediate)
- **Featured Engine (lecture demo):** GameMaker (GML script/text mode - callback to Lesson 5's GameMaker Visual demo)
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 8 projects (already saved) - no new assets needed this week

Agenda

0:00-0:05	Welcome back, Lesson 8 recap
0:05-0:15	The Payoff of Deferring UI
0:15-0:35	Score: Reusing the Hit-Detection Path (live demo in GameMaker)
0:35-0:50	Two Ways to Stay in Sync: Push vs. Poll
0:50-0:55	Introduce lab exercise, hand out engine-specific handouts
0:55-1:50	LAB: Add score tracking and a real on-screen UI to your project
1:50-2:00	Exit check / quick share-out

Levels for this lesson's exercise

- Level 1 (Minimal): destroying an enemy adds 10 points to score; score and lives are both visible on screen at all times.
- Level 2 (Complete): the score/lives readouts never show a stale or wrong number - they always match reality, immediately.
- Level 3 (Refined): style the HUD, add scoring variety, and/or tighten the separation between game logic and display code.

See: Lesson 9 Teacher's Script, Lesson 9 Prep Checklist, and the per-engine Student Handouts. Unity, Unreal, and 001 Game Creator involve genuinely new UI setup this week - budget extra circulation time for those clusters.